

EXECUTION COST AND THE LIMITS OF BACKTEST EVIDENCE

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SUMMARY

A backtesting engine was built to test whether an adaptive execution algorithm — one that varies its trading rate with price movement — beats a constant rate. Under implementation shortfall it is 20 bps worse, losing on 94% of 300 simulated markets. Under total P&L it is significantly BETTER in every regime. The two metrics rank the rules in opposite orders, and resolving that disagreement is the study's central result. A separate study shows parameter search manufactures ~13 points of profit that does not survive out of sample. Four measurement errors were found and corrected, and four claims retracted or narrowed.

METHOD

- **Backtesting:** event-driven replay with position tracking, costs, percentage-of-volume limits, and square-root market impact ($\propto \sqrt{Q/V}$).
- **Walk-forward:** rolling folds within one history. Abandoned — three folds gave no power and one contributed 78% of the reported effect.
- **Monte Carlo:** 300 paired paths, later re-run on a generator matched to volatility clustering, volume persistence and volume/volatility coupling measured on AAPL, KO and SPY.

RESULT 1 — THE TWO METRICS DISAGREE, AND THAT IS THE FINDING

Adaptive vs constant participation across six regimes: shortfall +17 to +29 bps (worse) in all six; P&L +0.4 to +0.8 pp (better) in all four tested, every interval excluding zero. Cost decomposition resolves it: adaptive trades 8% fewer shares at identical cost per share (Rs 0.208 vs 0.207) — its P&L edge is churn reduction, not execution skill. Component ablation locates the shortfall damage in one rule, which reduces order size on the cheapest days: 7,437 shares bought per favourable day versus 13,386 on the rest.

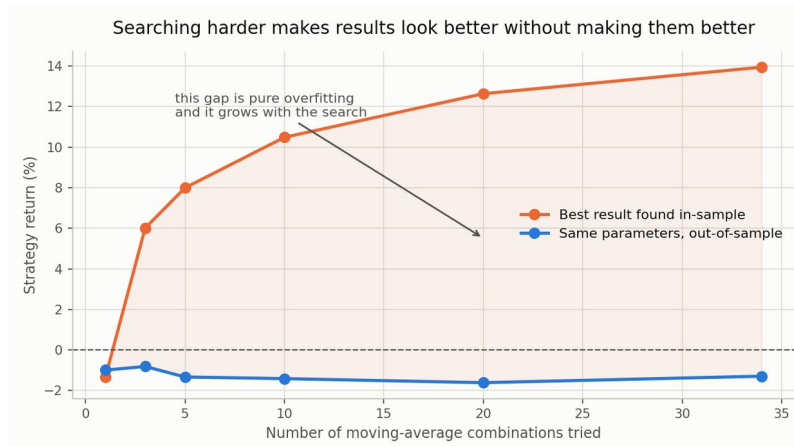
Corrected claim: adaptive achieves worse fill prices (+20 bps, robust) while producing better total P&L purely through reduced turnover. An execution metric alone would have reported the opposite of what reaches the account.

RESULT 2 — AN INVERTED RULE THAT WORKS

Reversing the logic — speed up when price moves in your favour, ease off when chasing — beats a constant rate by 8–11 bps in all six regimes, and improves P&L too. It is not simply faster: its average participation is 3.4%, slower than the 5% baseline, and a constant 3.4% rate costs 27–56 bps MORE. The 32–57 bps gap versus a speed-matched constant is genuine day-selection. Caveat: it holds ~6 points less of the target position, which is free only while the signal has no edge; the P&L advantage decays to zero as momentum rises.

RESULT 3 — PARAMETER SEARCH MANUFACTURES PERFORMANCE

Searching 34 moving-average window pairs on data built to contain no edge: best in-sample +11.4%, same parameters out-of-sample -1.6%. The illusion scales with search size (~1.3% at one candidate, +13.9% at 34). In-sample ranking predicts out-of-sample ranking at +0.003 — and stays there even in a control with a real edge planted, so knowing a strategy type works does not license tuning its parameters.



ERRORS FOUND AND CORRECTED

- Lookahead bias: order sizing used the day's realised volume, unknowable at decision time.
- Cost metric averaged per-order ratios, letting a 0.1-share order outweigh a 144,386-share order; true cost understated 6.7x.
- No market-impact model, so the metric silently rewarded pure speed.
- Warm-up asymmetry in the train/test split inflated the measured overfitting gap by 17%.

Four claims retracted or narrowed, including one where a component was judged by forecast accuracy rather than cost, and one where an execution metric was reported as the verdict.

WHAT THIS DOES NOT SHOW

All results are simulated. Real data calibrates the generator; it does not run the study. Real daily returns have autocorrelation ~0.00 — markets are approximately random walks in direction — so the strategy beat buy-and-hold on 0 of 12 real stock/window combinations. Missing: intraday dynamics, order-book depth, a calibrated impact constant, and any adversary reacting to order flow.